mux

July 30, 2025

[3]:
```
from main import peaks_vs_thresholds
import numpy as np
import matplotlib.pyplot as plt

%matplotlib widget
```

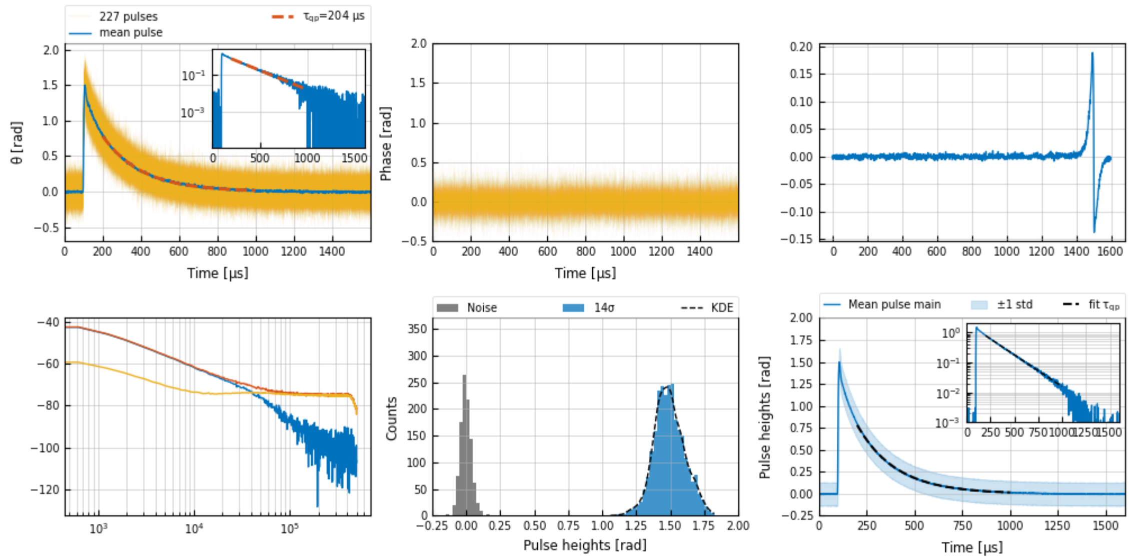
[19]:
```
kid = 5
pread = 113
file_type = 'vis'
pw = 1500
pw_offset = 100
filter_type = 'exp'
lifetime = 200
fit_tqp = [200, 1000]

dir_on = r"D:\Data\LT218Chip1_BF_20221103_MIR3_8\12KIDs mono on 3800␣
  ↪long\TD_Power"
chuncksize = 25
noise_mph = 10
pulse_mph = 14
nr_stds = [14]
nr_chuncks = [None]
```

[20]:
```
pulses = peaks_vs_thresholds(dir_on, kid, pread, file_type, noise_mph,␣
  ↪pulse_mph, pw, pw_offset, lifetime, nr_stds,
                             nr_chuncks, chuncksize, filter_type, fit_tqp)
```

```
Secondaries/Outliers/Total = 37/30/286
Optimal filter constructed with 227 pulses and 1000 noise segments
Analysed 300 out of 300 files
Secondaries/Outliers/Total = 437/370/3332

R = 6.04, Rsn = 15.68, 2624 pulses
tqp = 204 +- 0.5 us
```



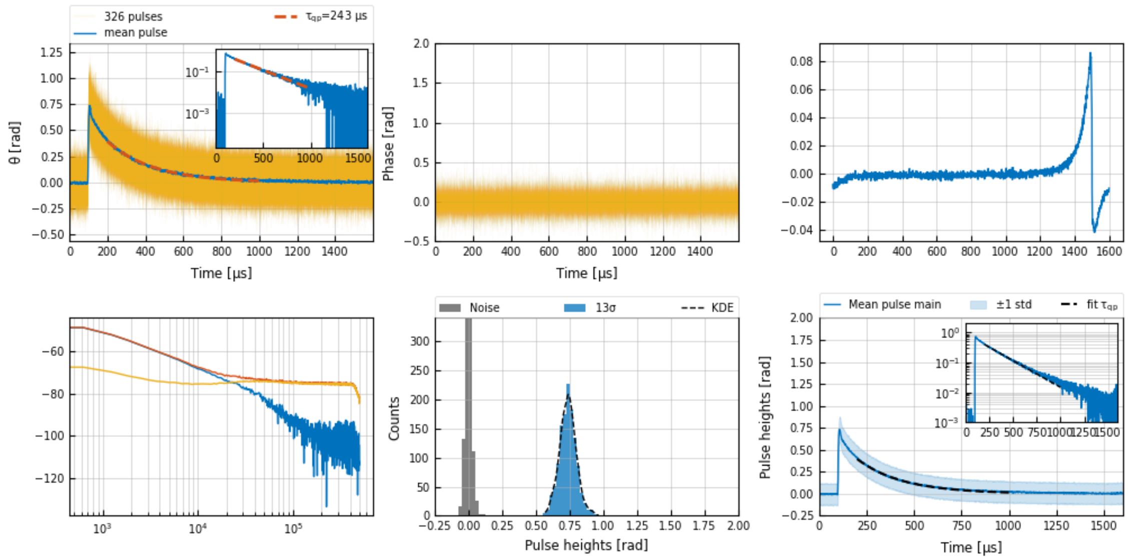
[21]:
```
kid = 5
pread = 113
file_type = 'vis'
pw = 1500
pw_offset = 100
filter_type = 'exp'
lifetime = 250
fit_tqp = [200, 1000]

dir_on = r"D:\Data\LT218Chip1_BF_20221025_MIR8_5\12KIDs mono off long\TD_Power"
chuncksize = 50
noise_mph = 5
pulse_mph = 13
doubles_fraction = 15
nr_stds = [13]
nr_chuncks = [None]
```

[22]:
```
pulses = peaks_vs_thresholds(dir_on, kid, pread, file_type, noise_mph,␣
 ↪pulse_mph, pw, pw_offset, lifetime, nr_stds,
                              nr_chuncks, chuncksize, filter_type, fit_tqp,␣
 ↪doubles_fraction=doubles_fraction)
```

```
Secondaries/Outliers/Total = 425/118/799
Optimal filter constructed with 326 pulses and 1000 noise segments
Analysed 200 out of 200 files
Secondaries/Outliers/Total = 1650/456/3042

R = 5.56, Rsn = 12.77, 1234 pulses
tqp = 243 +- 0.9 us
```



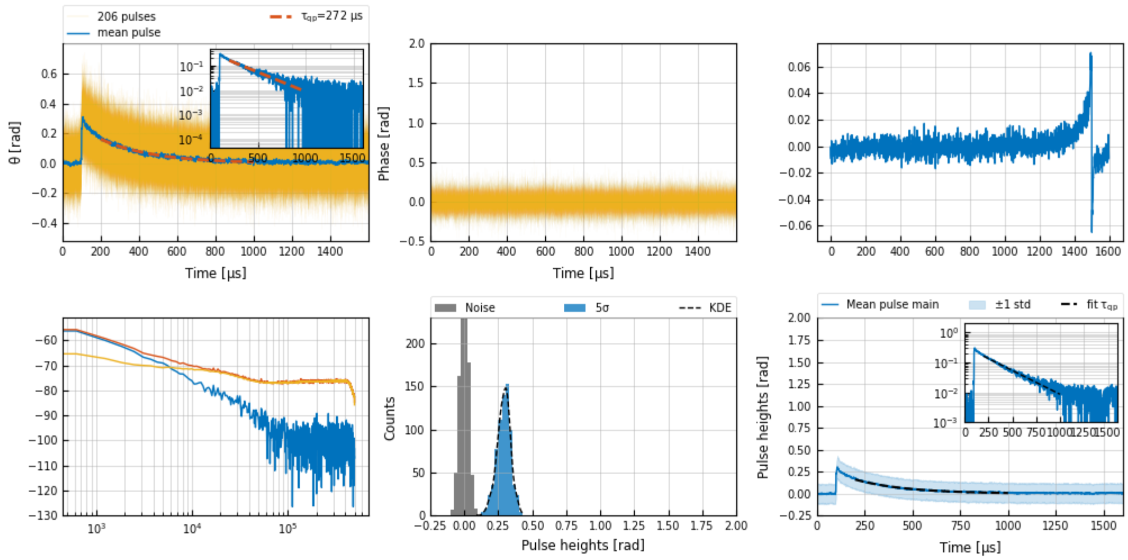
[23]:
```
kid = 5
pread = 113
file_type = 'vis'
pw = 1500
pw_offset = 100
filter_type = 'exp'
lifetime = 275
fit_tqp = [200, 1000]

dir_on = r"D:\Data\LT218Chip1_BF_20240116_MIR18_5\KID5_all_temps\180K"
chuncksize = 10
noise_mph = 5
pulse_mph = 5
doubles_fraction = 2
nr_stds = [5]
nr_chuncks = [None]
```

[24]:
```
pulses = peaks_vs_thresholds(dir_on, kid, pread, file_type, noise_mph,
    pulse_mph, pw, pw_offset, lifetime, nr_stds,
                              nr_chuncks, chuncksize, filter_type, fit_tqp,
    doubles_fraction=doubles_fraction)
```

```
Secondaries/Outliers/Total = 7/24/236
Optimal filter constructed with 206 pulses and 1000 noise segments
Analysed 40 out of 40 files
Secondaries/Outliers/Total = 30/124/860

R = 2.77, Rsn = 3.81, 720 pulses
tqp = 272 +- 2.6 us
```



[25]:
```
kid = 5
pread = 109
file_type = 'vis'
pw = 1500
pw_offset = 100
filter_type = 'exp'
lifetime = 320
fit_tqp = [200, 1000]

dir_on = r"D:
 ↪\Data\LT218Chip1_ADR_20240506_MIR24\12KIDs_3Pread_TD40s_1MHz_BB24K\TD_Power"
chuncksize = 20
noise_mph = 5
pulse_mph = 5
doubles_fraction = 5
nr_stds = [5]
nr_chuncks = [None]
binsize = 0.035
```

[26]:
```
pulses = peaks_vs_thresholds(dir_on, kid, pread, file_type, noise_mph,␣
 ↪pulse_mph, pw, pw_offset, lifetime, nr_stds,
                             nr_chuncks, chuncksize, filter_type, fit_tqp,␣
 ↪binsize=binsize, doubles_fraction=doubles_fraction)
```

```
Secondaries/Outliers/Total = 123/19/356
Optimal filter constructed with 225 pulses and 1000 noise segments
Analysed 40 out of 40 files
Secondaries/Outliers/Total = 255/42/725
```

```
R = 2.45, Rsn = 4.27, 448 pulses
tqp = 313 +- 0.7 us
```